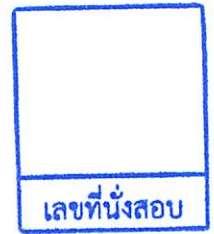


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of semester 1

Year: 2015

Subject: 392133 Physics III

Section: 15 - 16

Date: 3 December 2015

Time: 08.00 - 10.00

Name.....ID.....Field of Study.....

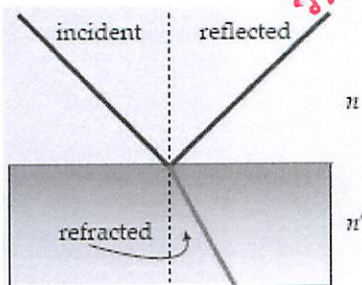
Instructions

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken inside the examination room.
 3. Express your answer in English. Other languages will be discarded.
 4. No calculators are allowed.
 5. Electronic communication devices are NOT allowed in the examination room.
 6. The examination has 7 page (including this page), 6 problems in the exam with the total scores of 60 points.
 7. Write all your answers on this examination sheet.
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1.1 [5 points] Two mirrors are at right angles (90°) to one another. A light ray is incident on the first at an angle of 30° with respect to the normal to the surface. What is the angle of reflection from the second surface?

1.2 [5 points] A light ray is partially reflected and partially refracted at a boundary between two media, the upper one having index of refraction n , the lower one having index of refraction n' , as shown in the figure. Prove that the reflected ray is perpendicular to the refracted ray when

$$n' = n \tan \theta_{\text{incident}}$$



Note: $\sin(A \pm B) = \sin A \cos B \pm \sin B \cos A$

2.1 [5 points] A concave mirror has a focal length of 20 cm. What is the position (in cm) of the resulting image if the image is inverted and four times smaller than the object?

2.2 [5 points] Suppose an object is placed 6.0 cm in front of a lens that has a focal length of 4.0 cm. Where is the image located? What are the magnification and the character of the image? Draw a ray diagram for a conclusion.

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3.1 [5 points] A contact lens is made of plastic with an index of refraction of 1.50. The lens has an outer radius of curvature of +2.0 cm and an inner radius of curvature of +2.5 cm. What are the focal length and the power of the lens?

3.2 [5 points] An object is placed 25 cm in front of a lens of focal length 20 cm. 80 cm past the first lens is a second lens of focal length 20 cm. How far past the second lens does the final image form?

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4.1 [5 points] A far-sighted student has a near point of 1.0 m. Calculate the focal length (in cm) of the glasses needed so the near point will be normal (25 cm).

4.2 [5 points] A 10-cm focal length converging lens is used to magnify small newspaper print 2.0 mm high. Calculate the height of the image (in mm) for the maximum magnification for a normal eye.

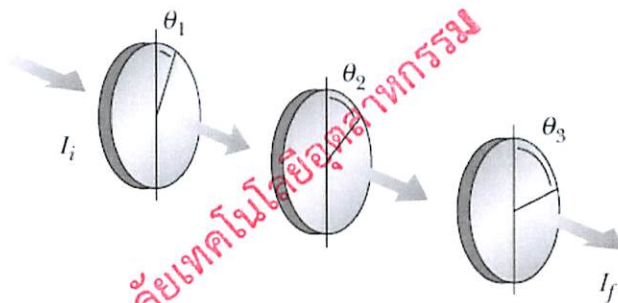
5.1 [4 points] A screen containing two slit d apart is L from the viewing screen. Light of wavelength λ falls on the slits from a distance source. Approximately how far apart will the dark interference fringes be on the screen?

5.2 [3 points] A slit of variable width illuminated by light of wavelength λ is widened until the first order diffraction minimum moves out to a distance infinitely far away from the brightest location in the central maximum. Determine the width of the slit in terms of the wavelength.

5.2 [3 points] A laser ($\lambda = 500 \text{ nm}$) is used to calibrate a diffraction grating. If the first-order maximum occurs at 30° , what is the line spacing, d ?

6.1 [5 points] A soap bubble appears green ($\lambda = 540 \text{ nm}$) at the point on its front surface nearest the viewer. What is the minimum thickness? Assume the refractive index of the soap bubble is 1.35.

6.2 [5 points] In Figure shown below, suppose that the transmission axes of the left and right polarizing disks are perpendicular to each other. Also, let the center disk be rotated on the common axis with an angular speed (ω).



Show that if unpolarized light is incident on the left disk with an intensity I_i , the intensity of the beam emerging from the right disk is

$$I_f = \frac{1}{16} I_i (1 - \cos 4\omega t)$$

Note: $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$ and, $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$

Hints: You can assume $\theta_1 = 0$, $\theta_2 = \omega t$ and $\theta_3 = 90^\circ$.